QP-based hammering task for a humanoid robot that leverages effective mass maximization for optimizing configuration at impact

Jimmy Vu†, Hélène Stefanell†, Ruud Erens†, Rafael Cisneros-Limón†, Mehdi Benallegue† and Abdelaziz Benallegue†

***Abstract*— Humanoid robots have the potential to relieve manual workers from wearing and exhausting tasks that may cause chronic joint damage over time. In this paper, we focus on a hammering task with our humanoid robot HRP-5P and we propose a method to hit a nail placed on a wood table with a given impact for a desired hitting velocity based on effective mass maximization and Quadratic Programming. Moreover, we implement a 5th order Bezier curve as the trajectory to attain the nail at any reachable pose.**

***Index Terms*— quadratic programming, hammering, humanoid robot, effective mass, end-effector trajectory**

I. Introduction

Manual workers in physically difficult and wearing industries, such as the construction industry, have a 50% higher risk of developing musculoskeletal disorders than people in other industries [10]. Inadequate working posture and repetitive movements have been showed to contribute to musculoskeletal disorders [6], which is the case for a hammering task. Furthermore, [6] demonstrated that fatigue affects elbow motion and shoulder injury affects both wrist and elbow motions during hammering. The study [4] showed that the increased moment exerted on joints during hammering can lead to more injuries, where a larger hammer causes a bigger increased moment. Unfortunately, larger hammers are often the preferred choice to accomplish such task. Indeed, as a result from [9] indicates, hammers themselves are one of the non-powered handtools that cause the most injury [3], especially to the fingers which were the most injured body part. Another cause of potential physical damage to workers is the position and orientation the hammering task has to be carried out in : [5] studied the optimal height to realize this task and concluded it should be between 25 cm and 35 cm below elbow height. However, it is not always possible to be at this height due to the variable working area in construction.

To relieve these workers from such joint-stressing activites, AIST developed the HRP-5P humanoid robot [8] which can take care of the wearing and exhausting tasks, such as carrying heavy loads or hammering a nail. The task of hammering a nail is the topic of this article wit the goal of hitting the nail and driving it into a wooden surface. We will use the mc_rtc open-source framework [1] and its Quadratic Programming (QP) solver to develop this task.

In this paper, The setting to complete this task is has a fixed hammer in one of the robots' grippers, a fully known position and orientation of the nail and a prefixed desired hitting velocity of the hammer tip on the nail. Section II recalls the robot dynamic model that we use, in Section III we propose a model for the nail-hammer impact, Section IV details the usage of QP in our work to maximize the impact on the nail, in Section V we explain the trajectory of the end-effector to the nail, simulations are presented in Section VI and finally we conclude our paper in Section VII.

II. Robot dynamic model

A. Robot dynamic equation

Let n be the number of Degrees Of Freedom (DOF) of the humanoid robot. The dynamic equation of the robot writes :

$$\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q},\dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{g}(\mathbf{q}) = \mathbf{u} + \sum_i \mathbf{J_i^T F_i} \tag{1}$$

where $\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}} \in \mathbb{R}^n$ are the configuration, configuration velocity and configuration acceleration of the robot respectively, $\mathbf{M}(\mathbf{q}) \in \mathbb{R}^{n\times n}$ is the symmetric mass matrix or inertia matrix of the robot, $\mathbf{C}(\mathbf{q},\dot{\mathbf{q}}) \in \mathbb{R}^{n\times n}$ is the Coriolis matrix, $\mathbf{g}(\mathbf{q}) \in \mathbb{R}^n$ is the vector of gravitational effects, $\mathbf{u} \in \mathbb{R}^n$ is the input vector of torques, $\mathbf{F_i} \in \mathbb{R}^6$ is the vector of the i-th external wrench (or generalized forces) i.e. forces and torques on each axis, $\mathbf{J_i} \in \mathbb{R}^{6\times n}$ is the absolute Jacobian of the point of application of $\mathbf{F_i}$ and $\{\cdot\}^\mathbf{T}$ is the matrix transpose operator.

B. Jacobian of the end-effector pose

Let $\mathbf{J} \in \mathbb{R}^{6\times n}$ be the Jacobian matrix of the end-effector pose defined such that :

$$\dot{\mathbf{x}} = \mathbf{J}(\mathbf{q})\dot{\mathbf{q}} \tag{2}$$

where $\dot{\mathbf{x}} \in \mathbb{R}^6$ is the task space velocity (translational and angular velocities) of the end-effector and the time derivative of the end-effector pose $\mathbf{x} \in \mathbb{R}^6$. The $\mathbf{J}$ matrix can be computed using forward kinematics :

$$\mathbf{J} = \frac{\partial \mathbf{x}}{\partial \mathbf{q}} \tag{3}$$

†CNRS-AIST JRL (Joint Robotics Laboratory), IRL3218, National Institute of Advanced Industrial Science and Technology (AIST), Tsukuba, Japan. Emails: jimmyvu2016@yahoo.fr, ruud.erens@solarteameindhoven.nl, {rafael.cisneros, mehdi.benallegue, abdelaziz.benallegue}@aist.go.jp

III. IMPACT MODELING

In this section, we propose a mathematical model of the impact between the hammer tip and the nail. This part is heavily inspired by W.J. Stronge's book "Impact Mechanics" [11] and his particle impact model.

A. Choice of the model

We have three main ways to solve this impact problem : i) a force-based model, ii) an energy-based model and iii) a momentum model.

1) Force-based model: The force-based model relies on Newton's second law :

$$\mathbf{F}^h = m_{e,\mathbf{n}} \frac{d\dot{\mathbf{x}}_\nu}{dt} \tag{4}$$

where $\dot{\mathbf{x}}_\nu \in \mathbb{R}^3$ is the translational part of $\dot{\mathbf{x}}$ defined in (2), $\mathbf{F}^h \in \mathbb{R}^3$ is the force applied to the nail by the hammer (the superscript "h" stands for "hammer") and $m_{e,\mathbf{n}}$ is the Effective Mass of the Robot (EMR) at the point of impact in the normal of the impact plane defined in (10). The issue with this model is the well-definedness of the acceleration of the end-effector, which is the time derivative of $\dot{\mathbf{x}}_\nu$. Indeed, due to the discontinuous velocity of the hammer tip at the instant of collision, the time derivative at t_c does not exist. Thus we will not use a force-based model.

2) Energy-based model: The energy-based model could be used with the issue of the discontinuity of the hammer tip velocity. However, during impact, the kinetic energy of the hammer tip is not fully transmitted to the nail : some of the energy is converted into heat, sound and deformation energy, which are difficult to estimate. The model is still usable, but we would either have to estimate the losses or we would have to neglect them making the model less precise.

3) Momentum-based model: The momentum based model does not have the issues of the force-based model and the energy-based model. With momentum, the microscopic interactions between the wood and the nail are taken into account through a scalar number e_* called *coefficient of restitution*, which measures the elasticity of the collision. Furthermore, another advantage of using this model is its ability to deal with discontinuous forces, which occur when hitting a nail with a hammer. That is why we use the momentum model in the following part of the paper.
More precisely, we will use a model based on particle impact. A schematic depicting the problem is proposed in Fig. 1. In this framework, an infinitesimal particle (i.e. with negligible radius) models the local deformations of the hammer tip and the nail.

B. Common tangent plane of collision

From [11], we define the common tangent plane of collision as the plane passing through the contact point at instant of collision and the normal unit vector to the collision $\mathbf{n} \in \mathbb{R}^3$ a vector orthogonal to that plane. Let $\mathbf{R}_{w\rightarrow nail} \in SO(3)$

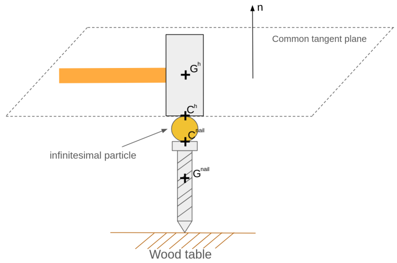

Fig. 1. Schematic of the collision using particle impact

be the rotation matrix of the nail with respect to the world frame w. Then the normal vector expressed in the world frame $\mathbf{n}_w$ is :

$$\mathbf{n}_w = \mathbf{R}^{\mathbf{T}}_{w\rightarrow nail}\mathbf{n} \tag{5}$$

C. Direct impact

Let $\mathbf{x}_\nu(t)$ and $\mathbf{c}^{nail}(t) \in \mathbb{R}^3$ be the contact points of the hammer tip and the nail expressed in the world frame w respectively, as illustrated by Fig. 1. At the instant of contact t_c, these points collide and thus have the same coordinates (neglecting the radius of the infinitesimal particle, see Fig. 1) :

$$\mathbf{x}_\nu(t_c) = \mathbf{c}^{nail}(t_c) \tag{6}$$

The nail is stationary before impact, i.e. from the initial time t_i to the instant of contact t_c its velocity is $\mathbf{0}$:

$$\dot{\mathbf{c}}^{nail}(t) = \mathbf{0} \qquad \forall t \in [t_i, t_c] \tag{7}$$

The velocity of the nail is given by the trajectory presented in Section V-C.
A direct impact is an impact that occurs when the velocity field in each colliding body is uniform and parallel to $\mathbf{n}$ [11]. Our goal with the hammering problem is to achieve a direct impact. We will assume that it is indeed the case, thus there will be no tangential sliding during impact.

D. Projected momentum of the hammer tip

To quantify an impact, we define the normal impulse i given by the hammer tip to the nail over an infinitesimal duration δt to be :

$$i \triangleq \int_{t_c}^{t_c+\delta t} \mathbf{n}^{\mathbf{T}}\mathbf{F}^h dt \tag{8}$$

To maximize impact, we want to maximize the impulse given by (8) which is equivalent to maximizing the projected momentum $P(t) \in \mathbb{R}$ of the hammer tip along the normal $\mathbf{n}$ defined as :

$$P(t) \triangleq m_{e,\mathbf{n}}(\mathbf{q})\dot{\mathbf{x}}_\nu^{\mathbf{T}}\mathbf{n} \tag{9}$$

where $m_{e,\mathbf{n}}(\mathbf{q}) \in \mathbb{R}^+$ is the configuration dependent EMR defined as :

$$m_{e,\mathbf{n}}(\mathbf{q}) \triangleq \left(\mathbf{n}^{\mathbf{T}}\mathbf{\Lambda}(\mathbf{q})\mathbf{n}\right)^{-1} \tag{10}$$

with $\mathbf{\Lambda} \in \mathbb{R}^{3\times3}$ defined as :

$$\mathbf{\Lambda}(\mathbf{q}) \triangleq \mathbf{J}_\nu(\mathbf{q})\mathbf{M}^{-1}(\mathbf{q})\mathbf{J}_\nu^{\mathbf{T}}(\mathbf{q}) \tag{11}$$

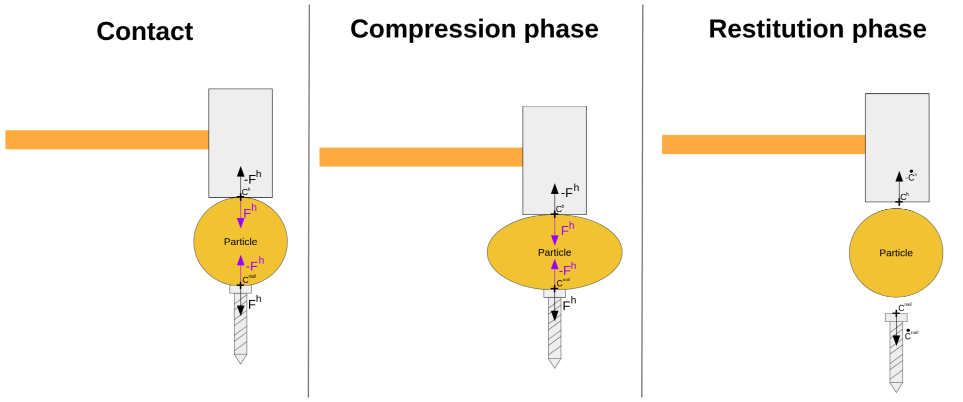

Fig. 2. Particle impact phases

with $\mathbf{J}_\nu \in \mathbb{R}^{3\times n}$ being the translational part of $\mathbf{J}$.
For a fixed given target normal velocity $\dot{\mathbf{x}}_\nu^{\mathbf{T}}\mathbf{n}$, we see that in order to maximize the projected momentum of the hammer tip, we have to maximize the EMR, which is explained in Section IV.

E. Particle impact

When the hammer tip collides with the particle, the particle will deform during the compression phase and store elastic strain energy. That energy will be released during the restitution phase and drive the hammer tip upwards and the nail downwards, see Fig. 2. From that, let us establish a mathematical model to assess the velocities of the robot and of the nail after impact. Let us define the effective mass of the collision $m \in \mathbb{R}^+$:

$$m \triangleq \frac{m_{e,\mathbf{n}} m_n}{m_{e,\mathbf{n}} + m_n} \tag{12}$$

where m_n is the mass of the nail which is very light compared to the EMR. Because of that, we have $m_n << m_{e,\mathbf{n}}$ and (12) becomes :

$$m = m_n \tag{13}$$

For ease of notation, let us introduce :

$$\dot{\mathbf{x}}_\nu^{t_c} \triangleq \dot{\mathbf{x}}_\nu(t_c) \tag{14}$$

From the particle impact model of Stronge [11], the post-impact velocity of the hammer tip $\dot{\mathbf{x}}_\nu^{+}$ is given by :

$$\dot{\mathbf{x}}_\nu^{+} = \left[1 - \frac{m}{m_{e,\mathbf{n}}}(1+e_*)\right]\dot{\mathbf{x}}_\nu^{t_c} \tag{15}$$

From the same model, the velocity of the nail after impact $\dot{\mathbf{c}}^{nail,+}$ is :

$$\dot{\mathbf{c}}^{nail,+} = \frac{m}{m_n}(1+e_*)\dot{\mathbf{x}}_\nu^{t_c} = (1+e_*)\dot{\mathbf{x}}_\nu^{t_c} \tag{16}$$

by the hypothesis given by (13).
Equation (15) shows that the hammer tip will be subject to a rebounce after impact and (16) shows that the velocity of the nail after getting hit depends on the pre-impact velocity of the hammer tip, which makes sense intuitively. The rebounce might destabilize the robot, but this problem is out of the scope of this paper.

IV. Optimization of the Effective Mass Using QP

In this section, we will describe how to maximize the EMR using QP.

A. Formulation of the problem to solve

For this part, our decision variable is the configuration acceleration $\ddot{\mathbf{q}}$. In order to make it appear on the expression of the effective mass, we differentiate the EMR twice with respect to time :

$$\dot{m}_{e,\mathbf{n}}(\mathbf{q},\dot{\mathbf{q}}) = \frac{dm_{e,\mathbf{n}}}{d\mathbf{q}}\frac{d\mathbf{q}}{dt} = \nabla m_{e,\mathbf{n}}^{\mathbf{T}}(\mathbf{q})\dot{\mathbf{q}} \tag{17}$$

$$\ddot{m}_{e,\mathbf{n}}(\mathbf{q},\dot{\mathbf{q}},\ddot{\mathbf{q}}) = \frac{d}{dt}\left\{\nabla m_{e,\mathbf{n}}^{\mathbf{T}}(\mathbf{q})\right\}\dot{\mathbf{q}} + \nabla m_{e,\mathbf{n}}^{\mathbf{T}}(\mathbf{q})\ddot{\mathbf{q}} \tag{18}$$

Let us give a mathematical justification to the method. From (18), we can reconstruct $m_{e,\mathbf{n}}$ by discrete time integration. Let Δt be the timestep of the controller. Then we have :

$$m_{e,\mathbf{n}}(\mathbf{q}(t+\Delta t)) = m_{e,\mathbf{n}} + \dot{m}_{e,\mathbf{n}}\Delta t + \ddot{m}_{e,\mathbf{n}}\frac{\Delta t^2}{2} \tag{19}$$

Taking $\arg\max_{\ddot{\mathbf{q}}}$ on both sides, the two first terms the of right hand side of the equation are discarded because they do not depend on $\ddot{\mathbf{q}}$ by (10) and (17) :

$$\arg\max_{\ddot{\mathbf{q}}} m_{e,\mathbf{n}}(\mathbf{q}(t+\Delta t)) = \arg\max_{\ddot{\mathbf{q}}} \ddot{m}_{e,\mathbf{n}} \tag{20}$$

Thus, in order to maximize $m_{e,\mathbf{n}}$ with respect to $\ddot{\mathbf{q}}$, we indeed have to maximize $\ddot{m}_{e,\mathbf{n}}$. Therefore, we want to solve :

$$\arg\max_{\ddot{\mathbf{q}}} \frac{d}{dt}\left\{\nabla m_{e,\mathbf{n}}^{\mathbf{T}}(\mathbf{q})\right\}\dot{\mathbf{q}} + \nabla m_{e,\mathbf{n}}^{\mathbf{T}}(\mathbf{q})\ddot{\mathbf{q}} \tag{21}$$

Again, because our decision variable is $\ddot{\mathbf{q}}$, we can drop the first term which does not depend on the configuration acceleration $\ddot{\mathbf{q}}$ and we are left with :

$$\arg\max_{\ddot{\mathbf{q}}} \nabla m_{e,\mathbf{n}}^{\mathbf{T}}(\mathbf{q})\ddot{\mathbf{q}} \tag{22}$$

which saves us the computation of the time derivative of the gradient.
As the QP in the mc_rtc framework uses a minimization problem, we can rewrite (22) to a minimization problem in the following way :

$$\arg\min_{\ddot{\mathbf{q}}} -\nabla m_{e,\mathbf{n}}^{\mathbf{T}}(\mathbf{q})\ddot{\mathbf{q}} \tag{23}$$

Let's call this problem the Effective Mass Maximization Task (EMMT). The goal of this task is to increase as much as possible the EMR at the next iteration of the controller without its value dropping down. However, we cannot assert that the result will be a global maximum because our problem is not convex. If we want to try to find a global maximum, we might need to use other methods, however these methods are generally much slower.

B. Linear QP constraints to be respected

Our linear constraints for the QP solver are described as i) joint limits, ii) joint velocity limits, iii) joint acceleration limits and iv) joint torque limits :

$$\mathbf{q_{LB}} \preceq \mathbf{q} \preceq \mathbf{q_{UB}} \tag{24}$$

$$\dot{\mathbf{q}}_{\mathbf{LB}} \preceq \dot{\mathbf{q}} \preceq \dot{\mathbf{q}}_{\mathbf{UB}} \tag{25}$$

$$\ddot{\mathbf{q}}_{\mathbf{LB}} \preceq \ddot{\mathbf{q}} \preceq \ddot{\mathbf{q}}_{\mathbf{UB}} \tag{26}$$

$$\boldsymbol{\tau}_{\mathbf{LB}} \preceq \boldsymbol{\tau} \preceq \boldsymbol{\tau}_{\mathbf{UB}} \tag{27}$$

where the subscripts $\mathbf{LB}$ and $\mathbf{UB}$ mean "Lower Bound" and "Upper Bound" respectively, $\boldsymbol{\tau} \in \mathbb{R}^n$ is the actual vector of joint torques and the $\preceq$ operator is the term-by-term comparison operator between two vectors of $\mathbb{R}^n$.

C. Implementation of the EMMT within a posture task

The posture task is a task where the robot has to maintain a given configuration $\mathbf{q}$. We can retrieve the expression of the cost function of the posture task from the complete cost function in our QP formulation as follows :

$$\arg\min_{\ddot{\mathbf{q}}} \frac{1}{2} W_p \|\ddot{\mathbf{q}} + \mathbf{b}\|^2 - W_m \nabla m_{e,\mathbf{n}}^{\mathbf{T}}(\mathbf{q})\ddot{\mathbf{q}} + C(\ddot{\mathbf{q}}) \tag{28}$$

where the first term represents the posture task, the second term is the EMMT and $C(\ddot{\mathbf{q}}) \in \mathbb{R}$ is the cost function associated to the other possible tasks that the robot has to realize (for instance a trajectory task). W_p, $W_m \in \mathbb{R}^+$ are the relative weights associated to the posture task and the EMMT respectively and $\mathbf{b} \in \mathbb{R}^n$ is a PD controller written as :

$$\mathbf{b} = K_p(\mathbf{q} - \mathbf{q_d}) + K_d(\dot{\mathbf{q}} - \dot{\mathbf{q}}_{\mathbf{d}}) - \ddot{\mathbf{q}}_{\mathbf{d}} \tag{29}$$

where K_p and K_d are the proportional (stiffness) and derivative (damping) gains respectively, $\mathbf{q_d}$, $\dot{\mathbf{q}}_{\mathbf{d}}$ and $\ddot{\mathbf{q}}_{\mathbf{d}} \in \mathbb{R}^n$ are the desired configuration, desired configuration velocity and desired configuration acceleration respectively.
The relationship between K_p and K_d is given by :

$$K_d = 2\sqrt{K_p} \tag{30}$$

To implement our EMMT task in mc_rtc, we will modify (28) to include the EMMT within the posture task and we finally have :

$$\arg\min_{\ddot{\mathbf{q}}} \frac{1}{2} W_p \|\ddot{\mathbf{q}} + \mathbf{b} - \frac{W_m}{W_p^2} \nabla m_{e,\mathbf{n}}(\mathbf{q})\|^2 + C(\ddot{\mathbf{q}}) \tag{31}$$

subject to (24), (25), (26) and (27).

V. Trajectory of the end-effector

In this section, we present how the end-effector of the robot reaches the nail with a given velocity.

A. B-Spline trajectory task

The primary task of the end-effector is to reach the nail with the given velocity. To do that, we use a continuous bezier curve as our trajectory, which is a B-Spline. Let us call that task the B-Spline trajectory task (BSTT).
A $(N+1)$-points bezier curve $\mathbf{\Gamma} : \mathbb{R} \to \mathbb{R}^3$, or N-th order bezier curve, is defined as :

$$\mathbf{\Gamma}(t) \triangleq \sum_{k=0}^{N} B_k^N \left(\frac{t - t_i}{T}\right) \mathbf{p}_k \quad \forall t \in [t_i, t_c] \tag{32}$$

where $[t_i, t_c]$ is the time interval on which $\mathbf{\Gamma}$ is defined, $t_c > t_i$, $\mathbf{p}_k \in \mathbb{R}^3$ is the constant k-th control point of the curve, $T = t_c - t_i$ and B_k^N is a Bernstein polynomial that acts as a weight [7], such that :

$$B_k^N \left(\frac{t - t_i}{T}\right) \triangleq \binom{N}{k} \left(\frac{t - t_i}{T}\right)^k \left(1 - \frac{t - t_i}{T}\right)^{N-k} \tag{33}$$

The first control point of the curve, $\mathbf{p_0}$, is the 3D initial position of the hammer tip $\mathbf{x}_\nu^{t_i} \in \mathbb{R}^3$ (the subscript "i" stands for "*initial*") and the last control point, $\mathbf{p_N}$, is the 3D position of the nail $\mathbf{c}^{nail} \in \mathbb{R}^3$, both with respect to the fixed world frame. Thus :

$$\mathbf{p_0} = \mathbf{x}_\nu^{t_i} \tag{34}$$

$$\mathbf{p_N} = \mathbf{c}^{nail} \tag{35}$$

These points have the property as acting as waypoints for the trajectory, which is not the case for the potential in-between points.

B. Curve constraints

To make the end-effector reach the desired velocity when hitting the nail, we constrain the curve by adding four control points $\mathbf{p_{c_1}}$, $\mathbf{p_{c_2}}$, $\mathbf{p_{c_{N-1}}}$ and $\mathbf{p_{c_{N-2}}} \in \mathbb{R}^3$. These added points count towards the degree of the curve. The expression of the added points are given by [2]:

$$\mathbf{p_{c_1}} = \mathbf{p_0} + \frac{T}{N} \boldsymbol{\kappa}_{\dot{\mathbf{x}}_\nu^{t_i}} \tag{36}$$

$$\mathbf{p_{c_2}} = 2\mathbf{p_{c_1}} - \mathbf{p_0} + \frac{T^2}{(N-1)(N-2)} \boldsymbol{\kappa}_{\ddot{\mathbf{x}}_\nu^{t_i}} \tag{37}$$

which are added just after $\mathbf{p_0}$, where $\boldsymbol{\kappa}_{\dot{\mathbf{x}}_\nu^{t_i}} \in \mathbb{R}^3$ are the initial linear velocity constraints and $\boldsymbol{\kappa}_{\ddot{\mathbf{x}}_\nu^{t_i}} \in \mathbb{R}^3$ are the initial linear acceleration constraints. Also :

$$\mathbf{p_{c_{N-1}}} = \mathbf{p_N} - \frac{T}{N-1} \boldsymbol{\kappa}_{\dot{\mathbf{x}}_\nu^{t_c}} \tag{38}$$

$$\mathbf{p_{c_{N-2}}} = 2\mathbf{p_{c_{N-1}}} - \mathbf{p_N} + \frac{T^2}{(N-1)(N-2)} \boldsymbol{\kappa}_{\ddot{\mathbf{x}}_\nu^{t_c}} \tag{39}$$

where $\boldsymbol{\kappa}_{\dot{\mathbf{x}}_\nu^{t_c}} \in \mathbb{R}^3$ are the linear final velocity constraints and $\boldsymbol{\kappa}_{\ddot{\mathbf{x}}_\nu^{t_c}} \in \mathbb{R}^3$ are the final linear acceleration constraints. Thus, if we add the curve constraints, we will have six control points in total, hence a 5^{th} order Bezier curve.

C. Hammer tip velocity and acceleration

The desired velocity $\dot{\mathbf{x}}_{\nu,d}$ and acceleration $\ddot{\mathbf{x}}_{\nu,d}$ of the hammer tip from the initial instant t_i to the instant of contact t_c expressed with respect to the world frame w are given by the first and second time derivative of (32) respectively :

$$\dot{\mathbf{x}}_{\nu,d}(t) = \dot{\mathbf{\Gamma}}(t) \quad \forall t \in [t_i, t_c] \tag{40}$$

$$\ddot{\mathbf{x}}_{\nu,d}(t) = \ddot{\mathbf{\Gamma}}(t) \quad \forall t \in [t_i, t_c] \tag{41}$$

Both the velocity and acceleration of the hammer tip are affected by the curve constraints (more precisely the added control points) defined in the last subsection.

D. Orientation of the hammer tip

In addition to reaching the nail, we have to orient the hammer tip such that the impact with the nail is *collinear*. Let $\mathbf{G}^h$, $\mathbf{G}^{nail} \in \mathbb{R}^3$ be the center of mass of the hammer tip and the center of mass of the nail respectively. A collinear impact is an impact where $\mathbf{G}^h$, $\mathbf{G}^{nail}$ and the contact points $\mathbf{x}$ and $\mathbf{c}^{nail}$ are aligned [11], see Fig. 1. We constraint two rotation DoFs : the roll and the pitch of the hammer tip, and we let the yaw be free. Indeed, we want to align three vectors : the vector normal to the contact surface of the hammer tip which we will call $\mathbf{n}^h \in \mathbb{R}^3$ expressed in the hammer tip frame, the $\mathbf{n}$ vector and finally the vector between the hammer tip and the contact point of the nail. The latter is already done by the path to the nail.
Let $\mathbf{R}_{w\rightarrow h}$ be the rotation matrix describing the rotation of the hammer tip from the world frame, $\boldsymbol{\epsilon} = \mathbf{n}^h - \mathbf{n}$ the error vector between the desired vector ($\mathbf{n}$) and the actual vector ($\mathbf{n}^h$), $\dot{\mathbf{x}}_\omega \in \mathbb{R}^3$ the angular velocity vector of the hammer tip in the world frame given by :

$$\dot{\mathbf{x}}_\omega = -\mathbf{R}_{w\rightarrow h}\left[\mathbf{n}^h\right]_\times \mathbf{J}_\omega \dot{\mathbf{q}} \tag{42}$$

where $\mathbf{J}_\omega \in \mathbb{R}^{3\times n}$ is the angular part of $\mathbf{J}$ and $[\cdot]_\times : \mathbb{R}^3 \rightarrow \mathbb{R}^{3\times 3}$ is an operator defined, for all $\mathbf{a} = (a_x, a_y, a_z)^{\mathbf{T}} \in \mathbb{R}^3$, as :

$$[\mathbf{a}]_\times = \begin{bmatrix} 0 & -a_z & a_y \\ a_z & 0 & -a_x \\ -a_y & a_x & 0 \end{bmatrix} \tag{43}$$

The time derivative of (42) is :

$$\ddot{\mathbf{x}}_\omega = -\dot{\mathbf{R}}_{w\rightarrow h}\left[\mathbf{n}^h\right]_\times \mathbf{J}\dot{\mathbf{q}} - \mathbf{R}_{w\rightarrow h}\left[\mathbf{n}^h\right]_\times \left(\mathbf{J}_\omega \ddot{\mathbf{q}} + \dot{\mathbf{J}}_\omega \dot{\mathbf{q}}\right) \tag{44}$$

Let us call this task the Vector Orientation Task (VOT). The cost function of the task is :

$$C_{VOT}(\ddot{\mathbf{q}}) = \|K_{p,VOT}\boldsymbol{\epsilon} + K_{d,VOT}\dot{\boldsymbol{\epsilon}} - \ddot{\mathbf{x}}_\omega(\ddot{\mathbf{q}})\|^2 \tag{45}$$

where $K_{p,VOT}$ and $K_{d,VOT} \in \mathbb{R}^+$ are its stiffness and damping respectively.

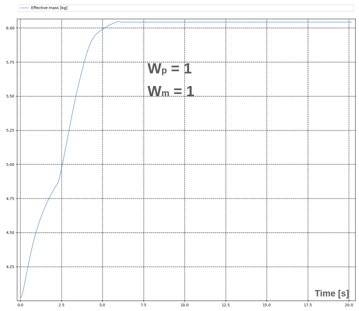

Fig. 3. Time evolution of the EMR with only a posture task active

VI. Simulations with the robot

In this section, we present the simulations made on our humanoid robot HRP-5P. Simulations were done using mc_mujoco and mc_rtc in torque control mode with a timestep $\Delta t = 0.002$s. The robot is equipped with a hammer in its left gripper and its floating base is fixed. The hammer used weighs 0.792 kg, has a length of 328 mm and its contact surface is circular with radius 15.5 mm. The nail is fixed and "floating" in the environment because of the current framework limitations, and has a length of 88.03 mm and its head has a radius of 6.95 mm. The tasks at play are the EMMT within the posture task defined in Section IV-A, the BSTT defined in Section V-A and the VOT defined in Section V-D. The gradient of the effective mass $\nabla m_{e,\mathbf{n}}$ is computed by backward difference. In (29), we set $K_p = 5$, thus by (30) we have $K_d = 4.47$.
The entire simulation goes as follows : i) the robot starts at its half-sitting posture ii) t_i is recorded and the robot starts the hammering motion until an impact is detected on the nail, then t_c is recorded at the same time, and finally iii) the robot goes back to its half sitting posture. An impact is detected on the nail thanks to a nail sensor plugin written in ROS2 and added to mc_rtc : there is an impact if the absolute value of one coordinate of the 3D force applied to the nail is at least greater than some value f_e which has been chosen to be 1N to cut off the very small "offset" forces of the simulator, as the magnitude of an impulse is much higher than this value. We first present the results of Section IV in Fig. 3 with $W_m = 1$ and $W_p = 1$ by isolating the posture task, i.e. we only have the posture task active and no other task active. We see that the EMR is growing over time until it converges to a maximum value. As expected from (31), increasing W_m induces a larger joint acceleration, thus the EMR converges faster and to a larger value, however it can lead to stability issues with the robot. Then, we use the already implemented *BSplineTrajectoryTask* and *VectorOrientationTask* of mc_rtc and we display the tracking of the theoretical trajectory given by the bezier curve (32), with the following curve constraints : $\boldsymbol{\kappa}_{\dot{\mathbf{x}}_\nu^{t_i}} = (0,0,0)^{\mathbf{T}}$, $\boldsymbol{\kappa}_{\ddot{\mathbf{x}}_\nu^{t_i}} = (0,0,0)^{\mathbf{T}}$, $\boldsymbol{\kappa}_{\ddot{\mathbf{x}}_\nu^{t_c}} = (0,0,0)^{\mathbf{T}}$ and especially

$$\boldsymbol{\kappa}_{\dot{\mathbf{x}}_\nu^{t_c}} = \mathbf{R}^{\mathbf{T}}_{w\rightarrow nail}\,{}^{nail}\dot{\mathbf{x}}_\nu^{t_c} \tag{46}$$

where ${}^{nail}\dot{\mathbf{x}}_\nu^{t_c} \in \mathbb{R}^3$ is the input velocity of the hammer tip at the instant of contact t_c expressed in the nail frame. The

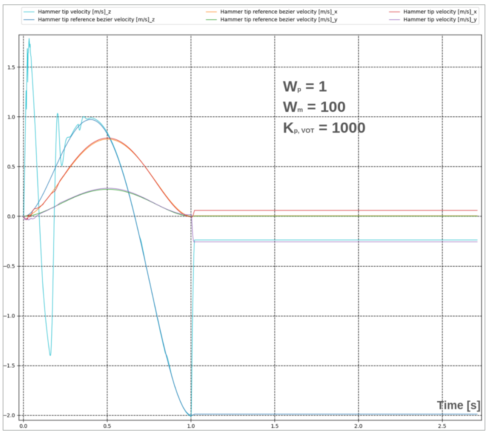

Fig. 4. Velocity tracking of the hammer tip, $K_{p,VOT} = 1000$

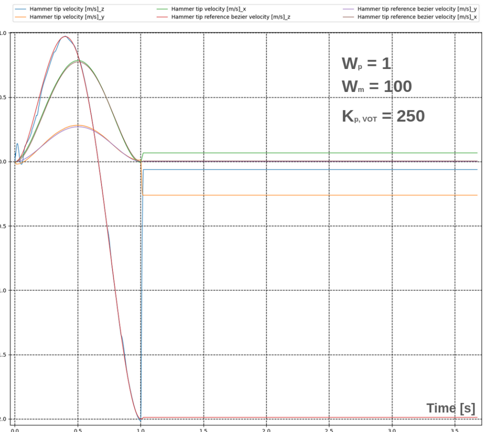

Fig. 5. Velocity tracking of the hammer tip, $K_{p,VOT} = 250$

nail is placed at $\mathbf{c}^{nail} = (0.75, 0.40, 0.90)^{\mathbf{T}}$ from the origin of w with the distances given in meters, and oriented as depicted by Fig. 1, i.e perpendicular to an horizontal plane. We set $T = 1$s, $W_{BSTT} = 100$, $W_{VOT} = 10$, $W_m = 100$, $W_p = 1$, ${}^{nail}\dot{\mathbf{x}}_{\nu}^{t_c} = (0, 0, -2.0)^{\mathbf{T}}$ m/s, the stiffness of the BSTT to $K_{p,BSTT} = 10000$ and the stiffness of the VOT to $K_{p,VOT} = 1000$ with their damping given by (30). We see the tracking of the hammer tip velocity in Fig. 4. We remark that the hammer tip properly reaches ${}^{nail}\dot{\mathbf{x}}_{\nu}^{t_c}$ until the sudden spike in velocity at the end, which is due to a collision with the nail. However, we also see some unwanted oscillations on the z coordinate : this is due to the high stiffness (1000) of the VOT. The final orientation error has been recorded to be 4.16 deg. We decrease the stiffness of the VOT to 250 in Fig. 5 and we record an orientation error of 11.88 deg. Increasing $K_{p,VOT}$ decreases the error between the desired and actual final hammer tip orientations at the cost of larger oscillations at the beginning.

VII. Conclusion

In this work, we presented a way to implement a hammering task on a humanoid robot. We used a momentum based model to estimate the dynamics of the robot and the nail after impact. Then, we proposed our method to maximize the EMR using QP and its gradient and we suggested a trajectory constructed from a constrained 5^{th} order bezier curve. Simulations showed that our EMMT is getting maximized and the trajectory given by the bezier curve is properly followed by the end-effector. Possible improvements include an estimation of the nail penetration in the wood along with a simulation of the dynamics of the nail, a better way to build our path and orientation to the nail or a better post-impact behavior that utilizes the post-impact velocity of the hammer tip to prepare for the next hit, as the robot only goes back to its initial posture for now. In the future, we want to work on the stability problem of the robot potentially induced by the post-impact rebounce of the hammer, minimizing the damage that the robot must endure in its joints during the hit to avoid damage as much as possible and detection of the nail via computer vision.

References

[1] The mc_rtc framework. URL: `https://jrl.cnrs.fr/mc_rtc/`.
[2] The ndcurves library. URL: `https://github.com/loco-3d/ndcurves/blob/devel/doc/curves.pdf`.
[3] F. Aghazadeh and A. Mital. “Injuries Due to Handtools: Results of a Questionnaire”. In: *Applied Ergonomics* 18.4 (Dec. 1987), pp. 273–278.
[4] Nilanthy Balendra and Joseph E. Langenderfer. “Effect of Hammer Mass on Upper Extremity Joint Moments”. In: *Applied Ergonomics* 60 (Apr. 2017), pp. 231–239.
[5] Yuan-Ho Chen and Yi-Lang Chen. “An Optimal Effective Working Height for Hammering Tasks”. In: *Work* 61.2 (Nov. 2018), pp. 181–187.
[6] Julie N. Côté et al. “Differences in Multi-Joint Kinematic Patterns of Repetitive Hammering in Healthy, Fatigued and Shoulder-Injured Individuals”. In: *Clinical Biomechanics* 20.6 (July 2005), pp. 581–590.
[7] David Salomon. *Computer Graphics & Geometric Modeling*.
[8] Kenji Kaneko et al. “Humanoid Robot HRP-5P: An Electrically Actuated Humanoid Robot With High-Power and Wide-Range Joints”. In: *IEEE Robotics and Automation Letters* 4.2 (Apr. 2019), pp. 1431–1438.
[9] Tadej Petrič et al. “Hammering Does Not Fit Fitts’ Law”. In: *Frontiers in Computational Neuroscience* 11 (May 2017).
[10] Scott P. Schneider. “Musculoskeletal Injuries in Construction: A Review of the Literature”. In: *Applied Occupational and Environmental Hygiene* 16.11 (Nov. 2001), pp. 1056–1064.
[11] W. J. Stronge. *Impact Mechanics*. 2nd ed. Cambridge University Press, 2018.